Designing N-of-1 Trials for a Biomarker-Treatment Interaction: What Drives Power, and How to Choose a Design

2026-10-02 18:12 PDT

Author. pmsimstats team

Purpose. This paper uses the closed-form power of the paired-difference interaction statistic (`docs/37`) to identify which characteristics of an N-of-1 or crossover design determine its power to detect a biomarker-by-treatment interaction, to search the space of blinded two-sequence designs for the best and most robust designs under compound-symmetry and AR(1) correlation, and to turn the results into a procedure for choosing a design.

Contents

Notation index and glossary 4
1. Summary 5
2. Framework 7
3. What drives the noise of the contrast 8
4. More on-off switches 10
5. Time between visits 11
6. Exhaustive design search 12
7. Open-label components 15
8. Blinded-discontinuation and crossover blocks 16
9. Monte Carlo checks 19
10. How to approach an optimal N-of-1 design 20
11. Limitations 21
12. Reproducibility 21
13. References 21

List of Tables

1 Notation index for the design optimization paper 4
2 Components of the contrast variance Var(Δ) under CS by design and path 8
3 Time part $a^\top Aa$ and Var(Δ) under AR(1) by design 9
4 Best attainable power by number of switches in the first sequence 10
5 Optimal visit spacing for the alternating design under AR(1) . . 12
6 Best design pattern, spacing and power in each condition 13
7 Design-spacing pairs with the highest worst-case relative power . 15
8 Best power by number of open-label visits, included in or excluded from the contrast 15
9 Power of four blinded block types after the open-label lead-in . . 17
10 Power of the published Hybrid design and three CO-block variants 18

List of Figures

1 Power against the time between visits for three blinded counterbalanced arrangements, by response half-life (light to dark, 0.25 to 4 weeks), under CS and AR(1) with 0%, 20% and 50% measurement error. Eight visits, so the trial lasts eight times the spacing. Points mark the spacing of highest power on the 0.25-6 week grid. 11

2 Every one of the 119 designs at its best spacing, against the number of switches in its first sequence, for each structure (rows) and half-life (columns). Dark open symbols: the crossover, ABAB and alternating reference designs. Orange: the best design in each condition. 14

Notation index and glossary

Added 2026-10-07. Notation follows `analysis/report/NOTATION.md` and `docs/37`. Symbols marked *local* are defined here. On 2026-10-07 the squared contrast-biomarker correlation was renamed from ρ^2 to $\rho^2_{\Delta B}$ (as in `docs/37`), the design-dependent slope from γ to $\beta_{bm:D}$, the mean-moderation strength from c_{bm} to β_{bm}, the biomarker SD from σ_B to σ_{bm}, the standardized biomarker from z to b, the measurement-error share from ν to ω, and the number of open-label visits from k to n_{OL}; γ, ν and k have other canonical meanings.

Table 1: Notation index for the design optimization paper

Symbol	Meaning	Status
N	total patients, 70, in two sequences of 35	canonical
t, w_t	visit index; visit time in weeks	canonical; w_t local
t_{sd}, $t_{1/2}$	time since discontinuation; response half-life (`docs/38`)	canonical
σ_{BR}, σ_{bm}	SDs of the drug response and of the biomarker	canonical
b	standardized biomarker	canonical
β_{bm}	mean-moderation strength, 0.25 here	canonical
c_{bm}	covariance-moderation correlation (mentioned only for its ceiling)	canonical
$\beta_{bm:D}$	slope of the contrast on the biomarker, the interaction estimand	canonical
ρ	within-factor correlation, 0.8 (between any two visits under CS, per week under AR(1))	canonical
h_t, $\bar{h}_{\mathrm{off}}$	share of the moderation present at visit t; its mean over off-drug visits	local
a_t, a	contrast weights, $1/n_{\mathrm{on}}$ on drug and $-1/n_{\mathrm{off}}$ off drug	local
n_{on}, n_{off}	on-drug and off-drug visits per patient	local
Δ	a patient's on-minus-off contrast	local
$\rho^2_{\Delta B}$	squared correlation between contrast and biomarker (the information)	local
C	response covariance over visits	local
K, A, A'	factor correlation; AR(1) time kernel; kernel with measurement error	local
ω	share of each factor's variance that is white measurement error	local
v	vector of factor SDs (constant expectancy)	local
η_t	design expectancy: 1 open label, 0.5 blinded (code `e`); written e_t before 2026-10-07	canonical
d	spacing between consecutive visits, weeks	local

Symbol	Meaning	Status
$f(d)$	large-sample information of the alternating design	local
n_{OL}	number of open-label lead-in visits	local

Glossary.

- **E9.** The paired-difference statistic of `docs/37`: each patient's on-drug mean minus off-drug mean, regressed on the biomarker; here stratified by sequence.
- **CS.** Compound symmetry: correlation 0.8 between any two visits of a factor.
- **AR(1) with measurement error.** Separable AR(1) correlation plus a white-noise share ω of each factor's variance.
- **Switch.** A change between on drug and off drug from one visit to the next within a sequence.
- **Pattern.** The on/off string of the first sequence (1 on, 0 off); the second sequence is its complement.
- **BD and CO blocks.** A blinded-discontinuation block (everyone on drug, paths differ in when the drug stops) and a crossover block.
- **Relative power.** Power of a design divided by the best power attainable in the same condition.

1. Summary

`docs/37` derived closed-form moments and power for the paired-difference interaction statistic (E9) in N-of-1 and crossover designs. This paper uses those expressions to ask which design characteristics determine power, and to search the design space for the best design. Without carryover the interaction slope is the same in every design, so a design affects power only through the noise of each patient's on-minus-off contrast and, under carryover, through how much of the biomarker's moderation survives at the off-drug visits. Every result below follows from that, and every number is a closed-form evaluation except where Monte Carlo is stated.

1. **What matters depends on how within-patient correlation behaves.** Under compound symmetry (CS), the published construct, correlation does not decay with time, so the arrangement of on and off visits is irrelevant: only the balance of on and off visits and the blinding matter. Under AR(1) correlation, the contrast cancels noise only between visits that are close in time, so interleaving on and off visits closely dominates (Sections 3 and 4).

2. **The Hybrid design's advantage is interleaving, and its open-label lead-in costs power.** Under CS, Hybrid is not the most powerful design: blinded balanced designs, including the crossover, match or beat it, and running the Hybrid schedule blinded raises its power from 0.72 to 0.80 (graded coupling; 0.663 to 0.739 under the decayed mean moderation of Section 2, corrected 2026-10-08, see Section 3). Under AR(1), Hybrid beats the crossover because its blinded-discontinuation visits sit next to on-drug visits and it switches three or four times; designs that alternate more often do better still (Section 3).
3. **More on-off switches help only under AR(1), and carryover limits them.** Under CS more switches never help and, with carryover, hurt: at a two-week response half-life the best one-switch design has power 0.731 and the best seven-switch design 0.559. Under AR(1) with 20% measurement error, the best seven-switch design has 0.435 at a one-week half-life against 0.221 for one switch (Section 4).
4. **There is an optimal time between visits under AR(1), roughly two to six response half-lives.** Under CS wider spacing is always better, up to a plateau reached when off-drug visits fall about two to five response half-lives after discontinuation, depending on the design. Under AR(1) the optimum is interior: close spacing cancels shared noise, wide spacing lets carryover wash out. For an alternating design it is about twice the half-life without measurement error and four to six times it when half the noise is measurement error (Section 5).
5. **The exhaustive search confirms the principles.** Over all 119 counterbalanced two-sequence designs with eight blinded visits, seven spacings, three correlation structures and three response half-lives, the best design is the crossover at the widest spacing under CS and, in five of the six AR(1) conditions, the alternating design at two to four weeks (the exception, 50% measurement error at a two-week half-life, favors a four-switch design). A near-alternating design at four-week spacing never falls below 78% of the best attainable power in any of the nine conditions (Section 6).
6. **Open-label visits cost power when they replace blinded ones.** Each open-label lead-in visit that takes the place of a blinded visit lowers power, under every structure. Whether the open-label visits should enter the on/off contrast depends on the correlation structure (Section 7).
7. **The closed form is confirmed by simulation.** Direct Monte Carlo of 41 designs, including the best design in every condition, agrees with the closed form within two standard errors in every case, and the further 96 design-conditions of the block comparison agree to within three (Sections 8 and 9).

2. Framework

Criterion. For a design with contrast weights a_t ($1/n_{\text{on}}$ on drug, $-1/n_{\text{off}}$ off drug), `docs/37` shows that power is governed by the per-patient squared correlation between the contrast and the biomarker,

$$\rho^2_{\Delta B} = \frac{(\beta_{bm:D}\,\sigma_{bm})^2}{\text{Var}(\Delta)}, \qquad \beta_{bm:D} = -\beta_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}(1-\bar{h}_{\text{off}}), \qquad \text{Var}(\Delta) = a^\top C a,$$

where $\bar{h}_{\text{off}}$ is the average share of the biomarker's moderation still present at the off-drug visits and C the response covariance. The interaction without carryover, $-\beta_{bm}\sigma_{BR}/\sigma_{bm}$, does not depend on the design. A design therefore matters through two quantities only: the noise $a^\top C a$ of the contrast, and the surviving moderation $\bar{h}_{\text{off}}$. The *information* reported below is the patient-weighted $\rho^2_{\Delta B}$.

(Corrected 2026-10-08: under mean moderation the biomarker shift adds $(\beta_{bm:D}\sigma_{bm})^2$ to the variance of Δ, and $a^\top C a$ is the residual variance of the contrast, net of that shift. The ratio displayed above is therefore the signal-to-noise ratio $r^2/(1-r^2)$, with $r^2 = (\beta_{bm:D}\sigma_{bm})^2/\{a^\top C a + (\beta_{bm:D}\sigma_{bm})^2\}$ the squared contrast-biomarker correlation, not the squared correlation itself. The two are monotone in each other, so no design ranking or power in this paper changes; the label $\rho^2_{\Delta B}$ is kept for continuity. See `analysis/report/compendium-mathematics.tex`, Section "Design efficiency and time adjustment", the criterion.)

Data-generating rule. Decayed mean moderation (`docs/37`, Section 5.6): the biomarker shifts the drug response by $\beta_{bm}\sigma_{BR} b\, h_t$, with $h_t = 1$ on drug and $2^{-t_{sd}/t_{1/2}}$ off drug, t_{sd} the time since discontinuation and $t_{1/2}$ the response half-life (`docs/38`). Its β_{bm} is on the same numeric scale as the covariance-moderation c_{bm} (NOTATION.md, rule 2), so the rule has the same slope as graded coupling at a matched value, but no ceiling, and the comparison of designs is not distorted by which effect sizes each design can represent. Under the covariance constructs, high-information designs have lower ceilings on c_{bm} (`docs/37`, Section 5.8), an artifact this choice avoids.

Correlation structures. Three, from `docs/37` Section 5.8 and an extension of it:

- **CS**, the published construct: correlation 0.8 between any two visits of a factor.
- **AR(1) with measurement error**: the separable form $K \otimes A'$ with time kernel $A'_{ts} = (1-\omega)\rho^{|w_t - w_s|} + \omega\,\mathbf{1}[t=s]$, $\rho = 0.8$ per week of separation, and a share ω of each factor's variance that is white measurement error, $\omega = 0.2$ or 0.5 (Figure 1 also shows $\omega = 0$).

The measurement-error share matters. Pure AR(1) has none, so visits placed very close together become nearly perfectly correlated, the contrast's noise vanishes, and the most powerful design would place visits as close together as possible for reasons that are not pharmacological. Symptom ratings carry measurement error, which sets a floor on the noise of any contrast; CS has an equivalent floor in its occasion-level term.

Designs and analysis. Eight post-baseline visits, $N = 70$, two counterbalanced sequences of 35 patients (an on/off pattern and its complement), blinded (expectancy 0.5) except where Section 7 adds an open-label lead-in, $\beta_{bm} = 0.25$. The analysis is E9 with sequence-specific intercepts, so that sequence and period differences, which E9's common intercept counts as noise (`docs/37`, Section 5.7), do not enter. Response half-lives of 0.5, 1 and 2 weeks.

3. What drives the noise of the contrast

Under CS, expanding $a^\top C a$ (`docs/37`, Section 4.1) shows four contributions. For E9 the weights sum to zero and the person-level component cancels, leaving (per path, verified):

Table 2: Components of the contrast variance Var(Δ) under CS by design and path

Design	Path	On / off visits	Balance term	Expectancy confound	Total Var(Δ)
Hybrid (published)	1, 2	5 / 3	17.5	3.2	44.1
	3, 4	4 / 4	16.4	5.0	45.0
Hybrid schedule, blinded	1, 2	5 / 3	17.5	0	38.3
	3, 4	4 / 4	16.4	0	35.9
OL+BDC	1	6 / 2	21.9	8.9	60.4
	2	5 / 3	17.5	12.8	56.4
CO, ABAB, alternating (blinded, 4 / 4)	all	4 / 4	16.4	0	35.9
CO with an open-label first period	both	4 / 4	16.4	20.0	64.2

- **Balance.** The balance term is proportional to $1/n_{\text{on}} + 1/n_{\text{off}}$, smallest at four and four; five and three costs 7%, six and two 33%.
- **Expectancy confounding.** Open-label visits are always on drug, so they make the on/off contrast partly an open-label against blinded contrast ($\sum_t a_t \eta_t \neq 0$). The person-level part of the expectancy factor then no longer cancels and enters the contrast, together with that factor's doubled standard deviation at open-label visits.
- **Arrangement does not matter.** Every blinded four-and-four design has the same variance, 35.9, however its on and off visits are ordered.

So under CS the Hybrid design's open-label lead-in costs power: blinded, the same schedule has power 0.80 instead of 0.72 (E9), and giving the crossover an open-label first period lowers its power from 0.82 to 0.56.

(Corrected 2026-10-08: these four powers, and the 0.72 to 0.80 of Section 1, item 2, were computed under graded covariance coupling at $c_{bm} = 0.25$, $t_{1/2} = 0$ (`02-design-characteristics.R`, `design-power.csv`), not under the decayed mean moderation that Section 2 fixes for this paper. Under the stated rule, with the stratified statistic of Section 6, the Hybrid design has power 0.663 published against 0.739 blinded, and the crossover 0.765 against 0.524 with an open-label first period. The direction of both effects, and their approximate size, are unchanged. See `analysis/report/compendium-mathematics.tex`, Section "Design efficiency and time adjustment", subsection "Contrast variance by design under CS and AR(1)".)

Under AR(1), with constant expectancy the separable variance is a product, $\text{Var}(\Delta) = (v^\top K v)(a^\top A a)$, and the time part $a^\top A a$ is where designs differ. (For the designs with open-label visits the product form does not hold exactly, which is why their $\text{Var}(\Delta)$ is not proportional to $a^\top A a$.) The time part is small when every on-drug visit has an off-drug visit close to it (per path, no measurement error, verified):

Table 3: Time part $a^\top A a$ and $\text{Var}(\Delta)$ under AR(1) by design

Design	Switches	$a^\top A a$	$\text{Var}(\Delta)$
CO	1-2	0.828	212.8
OL+BDC	2	0.649-0.664	183.9-192.7
Hybrid	3-4	0.361-0.449	107.3-126.8
ABAB, every 2.5 weeks	3-4	0.352	90.6
ABAB at the Hybrid visit times	3-4	0.299	76.7
Alternating every visit, every 2.5 weeks	7-8	0.165	42.3

This is the Hybrid design's real advantage over the crossover and OL+BDC designs: its weekly blinded-discontinuation visits sit next to on-drug visits, and it switches three or four times rather than once or twice. The same idea taken

further, an alternating design, has about a fifth of the crossover's contrast noise (42.3 against 212.8).

4. More on-off switches

From the exhaustive search of Section 6, the best power attainable with a given number of switches in the first sequence, at the best spacing (verified):

Table 4: Best attainable power by number of switches in the first sequence

Structure	$t_{1/2}$	1 switch	2	3	4	5	6	7
CS	0.5	0.765	0.765	0.764	0.764	0.763	0.763	0.763
CS	1	0.759	0.752	0.746	0.739	0.733	0.726	0.719
CS	2	0.731	0.703	0.675	0.645	0.616	0.584	0.559
AR(1), 20% error	0.5	0.268	0.386	0.453	0.512	0.510	0.522	0.547
AR(1), 20% error	1	0.221	0.282	0.339	0.389	0.397	0.404	0.435
AR(1), 20% error	2	0.210	0.239	0.269	0.291	0.297	0.301	0.314
AR(1), 50% error	0.5	0.257	0.315	0.347	0.378	0.381	0.385	0.400
AR(1), 50% error	1	0.243	0.275	0.305	0.326	0.332	0.340	0.352
AR(1), 50% error	2	0.230	0.247	0.259	0.265	0.261	0.258	0.257

Under CS, more switches never help, and once carryover is substantial they hurt: each switch puts an off-drug visit soon after a discontinuation, where moderation survives. Under AR(1), more switches help, because each brings on and off visits closer together, but the gain shrinks as the half-life lengthens and as measurement error grows; at a two-week half-life with 50% measurement error it has vanished (0.230 to 0.265, with the best at four switches).

5. Time between visits

With the number of visits fixed at eight, the spacing d between visits trades two effects against each other: on and off visits close together share more of the slowly varying noise, which the contrast cancels (correlation ρ^d); off-drug visits far from discontinuation retain less moderation ($2^{-d/t_{1/2}}$). Figure 1 shows power against spacing for three arrangements (verified, closed form).

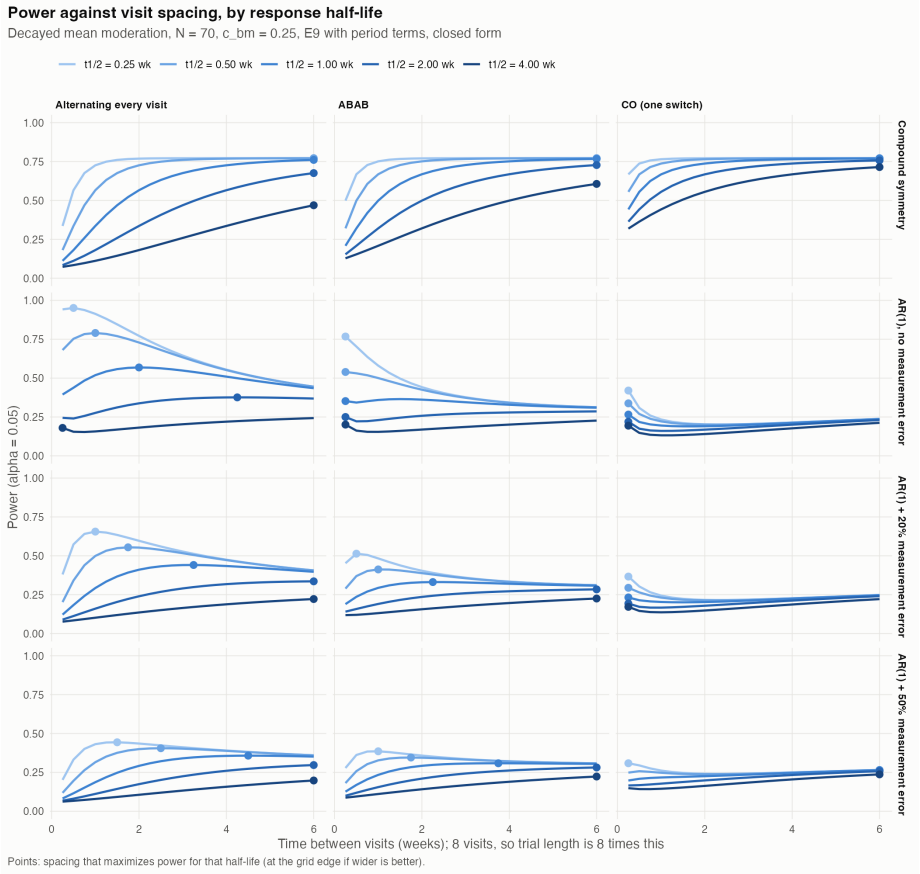


Figure 1: Power against the time between visits for three blinded counterbalanced arrangements, by response half-life (light to dark, 0.25 to 4 weeks), under CS and AR(1) with 0%, 20% and 50% measurement error. Eight visits, so the trial lasts eight times the spacing. Points mark the spacing of highest power on the 0.25-6 week grid.

Under CS there is no interior optimum. Correlation does not depend on spacing, so wider is always better, up to a plateau. The practical choice is the shortest spacing within 95% of the best power: for the alternating design 1.25, 2.25, 4.0 and 5.25 weeks at half-lives of 0.25, 0.5, 1 and 2 weeks, four to five

half-lives at the shorter half-lives and 2.6 at two weeks; for ABAB 1.0, 1.75, 3.25 and 4.75 weeks, about 2.4 to 4 half-lives; for the crossover 0.5, 1.0, 2.0 and 3.25 weeks, about two, because its placebo-first sequence carries no carryover and its off-drug blocks extend well beyond the discontinuation.

Under AR(1) there is an interior optimum. For the alternating design (verified):

Table 5: Optimal visit spacing for the alternating design under AR(1)

Measurement error	Optimal spacing at $t_{1/2}$ = 0.25 / 0.5 / 1 / 2 weeks	As multiple of $t_{1/2}$
none	0.5 / 1.0 / 2.0 / 4.25	about 2
20%	1.0 / 1.75 / 3.25 / beyond 6	about 3-4
50%	1.5 / 2.5 / 4.5 / beyond 6	about 4.5-6

Without measurement error the optimum has a closed form in the large-sample limit. Each off-drug visit is one spacing after a discontinuation, and the AR(1) time part of an alternating contrast is about $\frac{1}{2}(1-\rho^d)/(1+\rho^d)$, so information is proportional to

$$f(d) = \left(1 - 2^{-d/t_{1/2}}\right)^2 \frac{1+\rho^d}{1-\rho^d},$$

whose maximum, 0.45, 0.91, 1.88 and 4.34 weeks at the four half-lives, agrees with the exact grid optimum (verified). The more of the noise is measurement error, the less close spacing buys, and the further the optimum moves out. ABAB optima are shorter (0.5, 1.0 and 2.25 weeks at half-lives of 0.25, 0.5 and 1 with 20% error), because the second visit of each off-drug block is already two spacings after discontinuation. The best attainable power falls as the half-life lengthens: a drug with long carryover is intrinsically harder to study within patients.

6. Exhaustive design search

Space. Every on/off pattern of the eight visits with two to six on-drug visits, paired with its complement as the second sequence; a pattern and its complement are the same design, leaving 119 designs. Each was evaluated at spacings of 0.5, 1, 1.5, 2, 2.5, 3 and 4 weeks, under the three structures and three half-lives: 7,497 design evaluations.

Best design in each condition (pattern of the first sequence; verified):

Table 6: Best design pattern, spacing and power in each condition

Structure	$t_{1/2}$	Best pattern	Switches	Spacing (weeks)	Power
CS	0.5	11110000 (CO)	1	4.0	0.765
CS	1	11110000 (CO)	1	4.0	0.759
CS	2	11110000 (CO)	1	4.0	0.731
AR(1), 20% error	0.5	10101010 (alternating)	7	2.0	0.547
AR(1), 20% error	1	10101010	7	3.0	0.435
AR(1), 20% error	2	10101010	7	4.0	0.314
AR(1), 50% error	0.5	10101010	7	2.5	0.400
AR(1), 50% error	1	10101010	7	4.0	0.352
AR(1), 50% error	2	10011001	4	4.0	0.265

Under CS the optimum lies at the edge of the spacing grid, consistent with Section 5: wider would be better still, though by little (the CS plateau). Under AR(1) the optimum is interior for the shorter half-lives and moves to the edge as the half-life grows.

The landscape (Figure 2). Under CS most balanced designs cluster near the best, and the lower bands are the designs with unequal numbers of on and off visits, as Section 3 predicts: at a half-life of 0.5 weeks, power is 0.763-0.765 with four on-drug visits, 0.737-0.739 with three or five, and 0.643-0.645 with two or six (verified). Under AR(1) power generally rises with the number of switches (Section 4 gives the exceptions).

Robustness. The correlation structure and the response half-life of a real drug and outcome are uncertain, so a design that performs well across conditions is more useful than one that is optimal in one. The design-spacing pairs with the highest minimum, over the nine conditions, of power relative to that condition's best (verified):

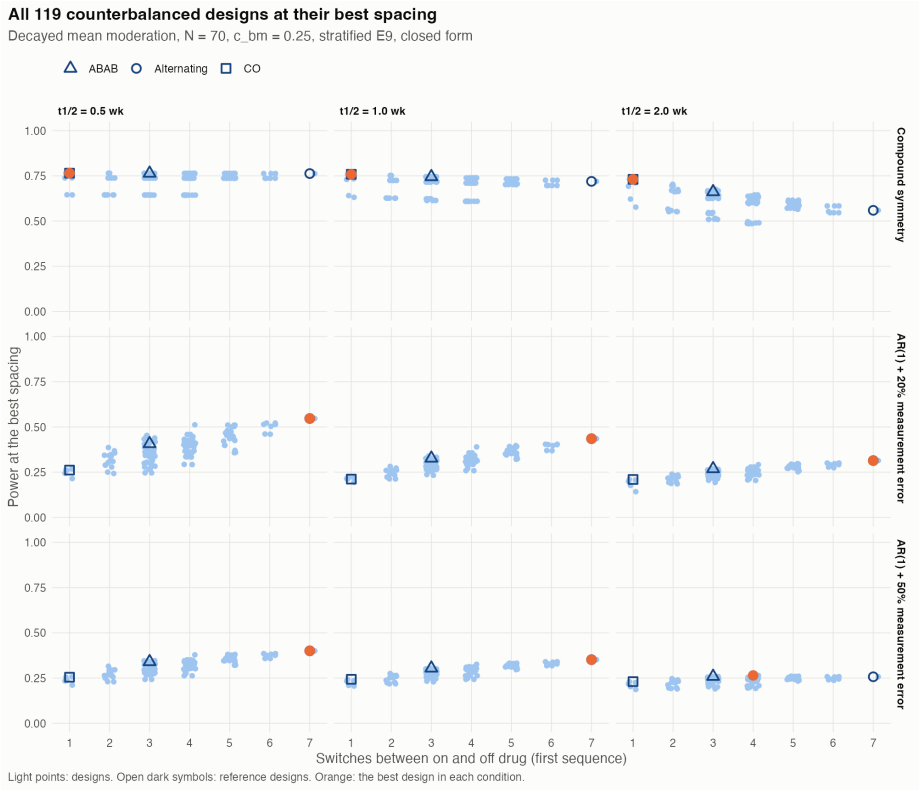


Figure 2: Every one of the 119 designs at its best spacing, against the number of switches in its first sequence, for each structure (rows) and half-life (columns). Dark open symbols: the crossover, ABAB and alternating reference designs. Orange: the best design in each condition.

Table 7: Design-spacing pairs with the highest worst-case relative power

Pattern (first sequence)	Spacing (weeks)	Worst relative power	Mean relative power
10101001 (tied with its mirror image 10010101)	4	0.78	0.92
10100101	4	0.77	0.91
10101010 (alternating)	4	0.77	0.94
10011001	3	0.76	0.88

The robust designs are alternating or nearly so, at wide spacing. They give up some power under CS with long carryover, where frequent switching is penalized, in exchange for most of the AR(1) gain.

7. Open-label components

Hendrickson et al.'s [1] Hybrid and OL+BDC designs begin with open-label visits, often needed for dose titration. In the data-generating model, expectancy η_t (1 at open-label visits, 0.5 at blinded ones) scales the placebo (expectancy) factor's mean and standard deviation. Because that factor is unrelated to the biomarker in the model, the higher placebo response during open-label visits does not bias the interaction slope; it changes only the noise. The search was repeated with $n_{\mathrm{OL}} = 0$ to 4 open-label on-drug visits for every patient, followed by counterbalanced blinded sequences over the remaining $8 - n_{\mathrm{OL}}$ visits, and two analyses: the open-label visits included in the on/off contrast, as E9 and the published analysis do, or left out of it. Best power over patterns and spacings (verified):

Table 8: Best power by number of open-label visits, included in or excluded from the contrast

Structure	$t_{1/2}$	$n_{\mathrm{OL}} = 0$	1, in / out	2, in / out	3, in / out	4, in / out
CS	0.5	0.765	0.722 / 0.701	0.679 / 0.644	0.605 / 0.551	0.547 / 0.479
CS	1	0.759	0.707 / 0.686	0.661 / 0.627	0.583 / 0.530	0.521 / 0.456
CS	2	0.731	0.638 / 0.616	0.582 / 0.549	0.491 / 0.443	0.421 / 0.366
AR(1), 20% error	0.5	0.547	0.463 / 0.487	0.355 / 0.422	0.280 / 0.361	0.237 / 0.291

Structure	$t_{1/2}$	$n_{\text{OL}} = 0$	1, in / out	2, in / out	3, in / out	4, in / out
AR(1), 20% error	1	0.435	0.370 / 0.374	0.306 / 0.328	0.251 / 0.274	0.215 / 0.228
AR(1), 20% error	2	0.314	0.257 / 0.258	0.217 / 0.228	0.181 / 0.192	0.160 / 0.163
AR(1), 50% error	0.5	0.400	0.358 / 0.352	0.311 / 0.311	0.262 / 0.261	0.231 / 0.220
AR(1), 50% error	1	0.352	0.314 / 0.305	0.278 / 0.271	0.238 / 0.227	0.210 / 0.194
AR(1), 50% error	2	0.265	0.228 / 0.217	0.208 / 0.197	0.179 / 0.165	0.161 / 0.144

(Corrected 2026-10-08: the entry for AR(1), 50% error, $t_{1/2} = 2$, two open-label visits excluded, was given as 0.196; it recomputes as 0.197, a quadrature difference in the last digit.)

- **Each open-label visit that replaces a blinded visit costs power**, under every structure and half-life: with the better of the two analyses, two open-label visits cost 0.09 to 0.15 under CS and 0.09 to 0.13 under AR(1) with 20% error. Part of the cost is the lost blinded visit; part is the open-label visits' noisier, confounded contribution.
- **Whether to include them in the contrast depends on the structure.** Under CS including them is better, plausibly because they add on-drug information that outweighs their expectancy noise (inferred). Under AR(1) with 20% measurement error leaving them out is better, plausibly because they come early in the trial, far from the blinded off-drug visits, and so add noise that the contrast cannot cancel (inferred). With 50% measurement error the two are close.

This comparison holds the total number of visits fixed. If an open-label titration phase is needed clinically, it can be added before the blinded visits rather than in place of them; left out of the contrast, it then costs no power, and included in it, its effect follows the pattern above. That variant was not evaluated.

8. Blinded-discontinuation and crossover blocks

The Hybrid design combines an open-label lead-in (weeks 4 and 8), a blinded-discontinuation (BD) block of four weekly observations (weeks 9-12), in which

everyone is on drug and the paths differ only in when the drug stops, and a crossover (CO) block of two observations four weeks apart (weeks 16 and 20), in which one sequence restarts the drug. Two questions: does a BD block perform differently from a CO block, everything else equal; and would the Hybrid design do better with a CO block like its BD block, four observations rather than two?

Like for like. After the common open-label lead-in, one block of four weekly observations (weeks 9-12), run in one of four ways, with two paths of 35 patients:

- **BD 10/9** (the Hybrid's): drug stopped after week 10 or after week 9 (on-on-off-off and on-off-off-off).
- **BD 11/9**: stopped after week 11 or week 9 (on-on-on-off and on-off-off-off).
- **CO**: on-on-off-off and off-off-on-on.
- **Alternating**: on-off-on-off and off-on-off-on.

Power with the open-label visits in the contrast (closed form, simulated in parentheses; verified; the AR(1) columns have 20% measurement error):

Table 9: Power of four blinded block types after the open-label lead-in

Block	CS, no carry-over	CS, $t_{1/2} = 0.5$	CS, $t_{1/2} = 1$	AR(1), none	AR(1), 0.5	AR(1), 1
BD 10/9	0.528 (0.525)	0.423 (0.416)	0.274 (0.281)	0.231 (0.224)	0.186 (0.186)	0.129 (0.131)
BD 11/9	0.450 (0.444)	0.326 (0.322)	0.199 (0.208)	0.219 (0.215)	0.162 (0.171)	0.110 (0.107)
CO	0.532 (0.527)	0.407 (0.413)	0.249 (0.251)	0.277 (0.274)	0.211 (0.215)	0.137 (0.137)
Alternating	0.532 (0.530)	0.335 (0.339)	0.176 (0.170)	0.356 (0.349)	0.222 (0.231)	0.125 (0.118)

- **Without carryover under CS the block type does not matter**: BD and CO blocks of the same length and balance have the same power (0.528 and 0.532), because CS does not see the order of the observations. Balance does matter: the BD variant whose paths stop at very different times (three on and one off against one on and three off) loses 0.08.
- **With carryover under CS the BD block is slightly better** (0.423 against 0.407 at $t_{1/2} = 0.5$; 0.274 against 0.249 at one week). Its off-drug observations run up to three weeks after the discontinuation, so less moderation survives (an average off-drug share of 0.33 against 0.38 at one week).
- **Under AR(1) the CO block is better** (0.277 against 0.231 without carryover). In its off-off-on-on sequence the off-drug observations lie between on-drug observations, the open-label visits before and the restart

after, so serially correlated noise is canceled on both sides; in the BD block every off-drug observation comes at the end.

The differences between the Hybrid's BD block and the CO block are at most 0.05 (0.004 to 0.046), small beside the effects of balance, spacing and the number of observations.

The Hybrid design with a four-observation CO block. Four designs, all with the open-label lead-in at weeks 4 and 8:

- **Published**: BD block of 4 weekly observations (weeks 9-12), CO block of 2 (weeks 16, 20); 8 visits, 20 weeks.
- **Equal visits**: BD block of 2 (weeks 10, 12), CO block of 4 every two weeks (weeks 14-20); 8 visits, 20 weeks.
- **CO weekly**: the published BD block and a CO block of 4 weekly observations (weeks 13-16); 10 visits, 16 weeks.
- **CO two-weekly**: the published BD block and a CO block of 4 observations every two weeks (weeks 14-20); 10 visits, 20 weeks.

Power (closed form, simulated in parentheses; verified):

Table 10: Power of the published Hybrid design and three CO-block variants

Design	CS, none	CS, 0.5	CS, 1	AR(1), none	AR(1), 0.5	AR(1), 1
Published	0.663 (0.666)	0.580 (0.571)	0.438 (0.435)	0.302 (0.297)	0.257 (0.267)	0.193 (0.194)
Equal visits	0.663 (0.673)	0.638 (0.640)	0.542 (0.551)	0.298 (0.300)	0.284 (0.285)	0.235 (0.236)
CO weekly	0.771 (0.772)	0.676 (0.674)	0.498 (0.502)	0.380 (0.385)	0.314 (0.314)	0.221 (0.228)
CO two-weekly	0.771 (0.776)	0.703 (0.698)	0.561 (0.580)	0.355 (0.351)	0.310 (0.303)	0.235 (0.234)

- **With the same eight visits and 20 weeks**, moving two observations from the BD block to the CO block changes nothing without carryover under CS (0.663 for both), and helps substantially with carryover (0.638 against 0.580 at $t_{1/2} = 0.5$, 0.542 against 0.438 at one week), because its off-drug observations fall two to eight weeks after a discontinuation, whereas most of the published design's fall one to three weeks after it: the average off-drug moderation share at one week falls from 0.245 to 0.134. Under AR(1) the reallocation helps with carryover and is neutral without it.

- **Adding two observations to the CO block**, the change suggested by extending it to four observations like the BD block, raises power in every condition, by 0.03 to 0.12, mostly through the extra observations and the balance they bring (the CS residual contrast variance $a^\top Ca$ falls from 44.58 to 34.27 at every half-life). (Corrected 2026-10-08: this was given as "from 46.9 to 36.3-36.7", which are path means of Var(Δ) including the mean-moderation shift at $t_{1/2} = 1$; the residual variance that determines precision is 44.58 for the published and equal-visit designs and 34.27 for the ten-visit designs. See `analysis/report/compendium-mathematics.tex`, Section "Design efficiency and time adjustment", subsection "Blinded-discontinuation and crossover blocks".) Weekly observations give the shorter trial and are better under AR(1) without carryover; two-weekly observations keep the 20-week length and are better under CS with carryover.
- **The open-label visits.** Under AR(1) leaving them out of the contrast helps the designs with a long CO block (0.440 against 0.380 for the weekly four-observation CO block without carryover), as in Section 7. With the equal-visit design it does not (0.263 against 0.298), because that design's blinded phase has fewer early on-drug observations to pair with its early off-drug observations (inferred).

In the decayed mean moderation used here, moderation is full at every on-drug observation, including immediately after a restart. Under the proportional moderation of `docs/38`, moderation would rebuild after a restart, which would count somewhat against CO blocks; that variant was not evaluated.

9. Monte Carlo checks

The stratified E9 statistic was simulated directly, 5000 trials of 70 patients per design, for the best design in each of the nine conditions, for the crossover, ABAB and alternating designs at 2.5-week spacing in every condition, and for the best two-visit open-label designs at a one-week half-life under both analyses: 41 designs. The draws come from the full joint distribution, including the AR(1) structures with measurement error, which no earlier check covered. Every design agrees with the closed form within two Monte Carlo standard errors (largest deviation 1.97, mean +0.35; verified). The closed form is, if anything, slightly conservative.

The block comparison of Section 8 was checked the same way: each of its eight designs simulated with 5000 trials under both structures and three half-lives, and the stratified E9 computed from the same trials with the open-label visits in and out of the contrast, 96 design-conditions. Agreement is within 2.95 standard errors (mean +0.11), with 3 of 96 beyond 2, fewer than the five or so expected by chance; the largest deviation is the alternating block under CS at $t_{1/2} = 0.5$ with the open-label visits left out (0.301 closed form, 0.320 simulated; verified).

10. How to approach an optimal N-of-1 design

The analysis suggests the following procedure; the principles are inferred from the closed form and confirmed for the designs simulated.

1. **State the two uncertain inputs.** The response half-life $t_{1/2}$ (`docs/38`: the persistence of benefit after discontinuation, not the drug's plasma half-life) and the within-patient correlation structure, in particular how fast correlation decays and how much of the variability is measurement error. Both can be estimated from pilot or historical data, and the design should be chosen to perform well across their plausible ranges.
2. **Blind the contrast, and balance it.** Equal numbers of on-drug and off-drug assessments per patient, with expectancy the same at on and off assessments. If open-label titration is necessary, add it before the blinded phase rather than in place of blinded assessments.
3. **Interleave if correlation decays.** Under AR(1)-like correlation, alternate frequently so that every on-drug assessment has an off-drug assessment nearby. Under CS-like correlation, arrangement does not matter and fewer, longer blocks are better under carryover.
4. **Space assessments against the response half-life.** Off-drug assessments should fall several half-lives after discontinuation: up to four or five under CS for alternating designs, two to six under AR(1) depending on the measurement-error share, fewer for long blocks in which most off-drug assessments are late.
5. **Counterbalance sequences and analyze with period terms.** Sequence and period differences otherwise enter the residual and cost power.
6. **Weight the contrast by the covariance.** E9 weights all on-drug and all off-drug visits equally, which is not optimal under AR(1); the best linear contrast among those whose weights sum to zero (so that it cancels the baseline level and any biomarker main effect) is proportional to $C^{-1}(h - \bar{h}_C \mathbf{1})$, with h the moderation pattern and $\bar{h}_C = \mathbf{1}^\top C^{-1} h / \mathbf{1}^\top C^{-1} \mathbf{1}$, which a correctly specified mixed model attains. (Corrected 2026-10-08: this was stated as C^{-1} times the moderation pattern, $C^{-1}h$, which omits the sum-zero projection. See `analysis/report/compendium-mathematics.tex`, Section "Design efficiency and time adjustment", subsection "The most efficient contrast".)
7. **Search the design space with the closed form.** The evaluation of one design is instantaneous, so the full space of sequences and spacings can be searched for the conditions of a given trial, and the shortlisted designs confirmed by simulation, as in Sections 6, 8 and 9.
8. **Choose block lengths and spacing, not block labels.** Blinded discontinuation and crossover blocks of the same length perform alike; what matters is how many on-drug and off-drug observations a block contributes and how long after a discontinuation its off-drug observations fall (Section 8).

11. Limitations

- **Fixed visit count and sample size.** Eight post-baseline visits, $N = 70$, two sequences of equal size. Trial length therefore varies with spacing; a fixed trial length would trade visits against spacing, a different question. Unequal spacing within a trial was not searched.
- **The data-generating rule.** Decayed mean moderation with anchored decay. Under the covariance constructs, high-information designs have lower c_{bm} ceilings and could not represent the same effect sizes; under proportional moderation (`docs/38`), on-drug moderation would also build up during titration.
- **The correlation structures.** CS and separable AR(1) with $\rho = 0.8$ per week and 0%, 20% or 50% measurement error. Which describes real symptom data is an empirical question that decides the ranking of designs.
- **The analysis.** Stratified E9 with equal weights. A mixed model with the correct covariance would gain somewhat, most under AR(1).
- **Expectancy.** The placebo factor is unrelated to the biomarker in the model. If expectancy effects were themselves moderated by the biomarker, open-label phases would bias the interaction, not only add noise.
- **Not all designs simulated.** The search is closed form; 41 designs were simulated, all agreeing.

12. Reproducibility

```
Rscript analysis/scripts/quick-sim/carryover-closed-form/02-design-characteristics.R
Rscript analysis/scripts/quick-sim/carryover-closed-form/03-optimal-spacing.R
Rscript analysis/scripts/quick-sim/carryover-closed-form/04-design-search.R \
  --mc-reps 5000 --cores 6
Rscript analysis/scripts/quick-sim/carryover-closed-form/05-block-comparison.R \
  --mc-reps 5000 --cores 6
```

Run from the repository root. All four scripts evaluate the definitions of `01-e9-closed-form.R` (`docs/37`). Outputs are written to `analysis/data/quick-sim/carryover-closed-form/` (design structure and spacing tables) and its `design-search/` subdirectory (the full search, best designs, robustness ranking, open-label search and Monte Carlo check), and the figures to `docs/figures/39-fig1-optimal-spacing.png` and `docs/figures/39-fig2-design-landscape.png`.

13. References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in*

Digital Health 2020; 2:13. doi:10.3389/fdgth.2020.00013. Code: `github.com/rchendrickson/pmsimstats`, commit `58b32a9`.
2. pmsimstats team. `docs/37-carryover-power-closed-form.md` (closed-form moments, the factor-by-time decomposition and the validation of the framework).
3. pmsimstats team. `docs/38-pharmacodynamic-carryover-model.md` (the response half-life and proportional moderation).
4. pmsimstats team. `docs/36-hendrickson-pd-and-carryover.md`.